

Question Label : Multiple Choice Question

A project earns a total of 50% return in two years. If annual return in the first year is 30%, then what is the annual return in the second year? (Round off to nearest integer)

Options :

6406533355893. ✖ 30%

6406533355894. ✖ 25%

6406533355895. ✖ 20%

6406533355896. ✔ 15%

Question Number : 271 Question Id : 640653993595 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the annualized rate of return of an investment of 100 units that promises to return 200 units in 10 years?

Options :

6406533355897. ✖ 10%

6406533355898. ✔ 7.18%

6406533355899. ✖ 5.97%

6406533355900. ✖ 3.76%

Question Number : 272 Question Id : 640653993596 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A financial advisor tells you that you can become a millionaire if you start investing early. You start to deposit 25,000 units every year in a savings account which earns 8% annual interest. What is the earliest year at which your account balance reaches a million units? (Assume you don't make any withdrawals in between)

Options :

6406533355901. ✖ 40

6406533355902. ✖ 25

6406533355903. ✔ 19

6406533355904. ✖ 15

LLM

Section Id :

64065369323

Section Number :

14

Section type :

Online

Mandatory or Optional :

Mandatory

Number of Questions :

10

Number of Questions to be attempted :	10
Section Marks :	40
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653146271
Question Shuffling Allowed :	No

Question Number : 273 Question Id : 640653993597 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : LARGE LANGUAGE MODELS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533355905. ✓ YES

6406533355906. ✗ NO

Question Number : 274 Question Id : 640653993598 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

General Note:

• If you find the given information is insufficient in any of the Numerical Answer Type (NAT) questions, enter -1 as the answer.

• In all your calculations, take at least two digits. For example, if the intermediate result is 1.860875, then take it as 1.86 to the next step.

Options :

6406533355907. ✓ Instructions has been mentioned above.

6406533355908. ✗ This Instructions is just for a reference & not for an evaluation

Sub-Section Number :

2

Sub-Section Id :

640653146272

Question Shuffling Allowed :

Yes

Question Number : 275 Question Id : 640653993604 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Teacher forcing for training an transformer model is:

Options :

6406533355913. ✖ mandatory.

6406533355914. ✔ optional.

optional but preferred

Question Number : 276 Question Id : 640653993616 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Choose the correct statements regarding transformer architecture:

Options :

6406533355937. ✖ Residual connections are practically optional in a transformer model with 10 encoder layers and 10 decoder layers. α

6406533355938. ✔ Having multiple heads helps in capturing different relationships between input tokens. α

6406533355939. ✖ One hot encoding is a good choice for position encoding α

6406533355940. ✖ Batch normalization can be used instead of layer normalization. α

6406533355941. ✖ None of these. α $\checkmark$

sparse matrix

Sub-Section Number :

3

Sub-Section Id :

640653146273

Question Shuffling Allowed :

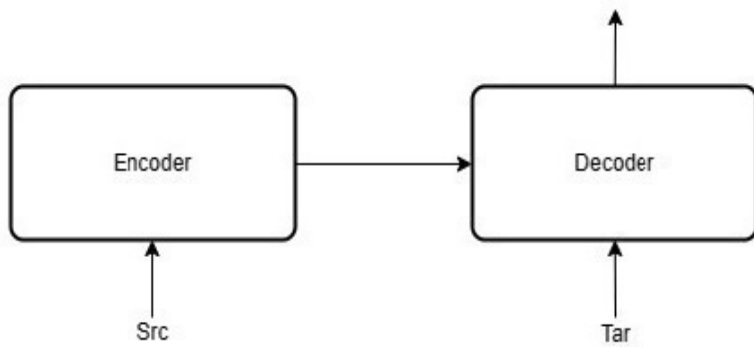
No

Question Id : 640653993599 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (277 to 280)

Question Label : Comprehension

Consider the transformer-based encoder-decoder model used for machine translation with the following configuration (except the source and target



vocabulary size, all other configuration values are the same for the encoder and decoder). The size of the vocabulary includes all the special tokens as well.

1. Source vocabulary size, $|V_s|$: 500
2. Target vocabulary size, $|V_t|$: 750
3. The context length, T : 128
4. The embedding dimension, $d_{embed} = d_{model}$: 64
5. The projection dimensions, $d_q = d_k = d_v$: 16
6. The number of heads, n_h : 4
7. The hidden dimension of FFNN, d_{ff} : 256
8. Uses layer norm and residual connection

We pass the sample from the source language “I enjoyed the movie Transformer” to the encoder and its translation “Naan transformer padaththai rasithen” to the decoder. All the learnable parameters are randomly initialized. The model uses the teacher-forcing approach throughout the training process.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 277 Question Id : 640653993600 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Assume the model has two layers ($N = 2$). Calculate the total number of parameters in the model (excluding the embedding layer and output layer). Moreover, no bias was added to the neuron in

$$enc = 2$$

$$dec = 2$$

the FFNN layers.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

230656

Question Number : 278 Question Id : 640653993601 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

How many parameters does the output layer have?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

48000

$d_{model} \times Target$

$$64 \times 750 = 48000$$

Question Number : 279 Question Id : 640653993602 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

How many parameters does the embedding layer have?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

32000

$d_{model} \times source$

$$64 \times 500 = 32000$$

Question Number : 280 Question Id : 640653993603 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

At time step $t = 1$, the prediction probability for the words "Naan", "transformer" and "padaththai" are 0.55, 0.15 and 0.2, respectively. What is the probability for the word "rasithen"? If you think, the given information is insufficient enter -1

Response Type : Numeric

Evaluation Required For SA : Yes

-1

we try because
0.1 probab. will

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-1



have distribⁿ over
186 words.

Question Id : 640653993617 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (281 to 284)

Question Label : Comprehension

Consider following input matrix to a single neural network layer:

$$\mu = 2$$
$$\sigma^2 = \frac{2}{3}$$

$$X = \begin{bmatrix} \downarrow & \downarrow & \downarrow \\ 1 & 2 & 3 \\ 4 & 7 & 11 \\ -3 & 5 & -2 \end{bmatrix}$$

$$x_i = \frac{1 - 2}{\sqrt{2/3}} = \frac{-1}{\sqrt{2/3}}$$
$$= \frac{-1}{\sqrt{2/3}}$$

It has 3 samples with 3 features each i.e. batch size is 3.
Each column represents a sample. Left most column represents
the first sample and topmost row represents the first feature vector.

Note: assume $\gamma = 1$, $\beta = 0$ and $\epsilon = 10^{-6}$ for both batch and
layer normalization, if not mentioned explicitly. Use standard
scaling.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 281 Question Id : 640653993618 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Say matrix F is computed by applying batch normalization on X , what will be sum of every element
in first/topmost row of F ?

Response Type : Numeric

Evaluation Required For SA : Yes

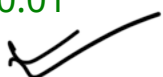
Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0 to 0.01



Question Number : 282 Question Id : 640653993619 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Say matrix F is computed by applying layer normalization on X , what will be sum of every element in first/leftmost column of F ?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0 to 0.01

$$\mu = 1 + 4 - 3 = 2/3 = 0.6$$
$$\sigma^2 = 0.16 + 12.96 + 3.71$$
$$= 6.29$$

$$\begin{array}{r} 0.4 \\ \sqrt{6.29} \\ \hline 3.6 \\ \sqrt{6.29} \\ \hline -3.6 \\ \sqrt{6.29} \end{array}$$

Question Number : 283 Question Id : 640653993620 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Say σ_1 , σ_2 and σ_3 were the standard deviations used to compute batch normalization of X .

What is $\sum_{i=1}^3 \sigma_i$?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

8.95 to 9.05

$$\mu_1 = \frac{2}{3}$$
$$\sigma_1^2 = \left(1 - \frac{2}{3}\right)^2 = \frac{1}{9}$$
$$\left(4 - \frac{2}{3}\right)^2 = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$
$$\left(-3 - \frac{2}{3}\right)^2 = \left(-\frac{11}{3}\right)^2 = \frac{121}{9}$$
$$= \frac{222}{9} / 3$$
$$\sqrt{\sigma_1^2} + \sqrt{\sigma_2^2} + \sqrt{\sigma_3^2}$$

Question Number : 284 Question Id : 640653993621 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Say σ_1 , σ_2 and σ_3 were the standard deviations used to compute layer normalization of X . What is $\sum_{i=1}^3 \sigma_i$?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :



Sub-Section Number :

4

1

Sub-Section Id :

640653146274

Question Shuffling Allowed :

No

Question Id : 640653993611 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (285 to 288)

Question Label : Comprehension

Consider a vocabulary $\mathcal{V}$,

$\mathcal{V} = ([\text{start}], \overset{0}{\text{breaking}}, \overset{1}{\text{ground}}, \overset{2}{\text{a}}, \overset{3}{\text{is}}, \overset{4}{\text{seismology}}, \overset{5}{\text{study}}, \overset{6}{\text{experience}}, \overset{7}{\text{earthquake}}, \overset{8}{[\text{end}]}]$.

Assume that we have a pre-trained GPT model for text generation and the first input token to the model is always the special token [start]. The prediction probabilities outputted by the model are given below.

over $\rightarrow$

					is	sci	study		
					↓	↓	↓		
$\hat{Y} =$	0.04	0.01	0.14	0.07	0.09	0.41	0.08	0.05	0.08
	0.10	0.03	0.06	0.16	0.43	0.01	0.06	0.09	0.01
	0.06	0.08	0.28	0.29	0.03	0.08	0.04	0.08	0.04
	0.11	0.03	0.48	0.02	0.01	0.07	0.06	0.14	0.07
	0.09	0.29	0.04	0.01	0.04	0.03	0.25	0.05	0.15
	0.14	0.22	0.13	0.08	0.01	0.03	0.06	0.23	0.08

The columns of the matrix are in the same order as the words listed in the vocabulary set above

The zeroth row of the matrix is the output probability distribution by the model given the special [Start] token as input.

Following the Greedy Search decoding strategy, the subsequent rows give the conditional probability distribution conditioned over the previous tokens. If you think the given information is insufficient for any sub-questions, then enter -1 as your answer

Based on the above data, answer the given subquestions.

Sub questions

for
study is easy
0.41 x 0.43
x 0.09
x 0.06

Question Number : 285 Question Id : 640653993612 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the probability of the predicted word equal to "Seismology" given the "[Start]" token as the initial input $P(y_1 = \text{Seismology} | y_0 = [\text{Start}])$?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.41



Question Number : 286 Question Id : 640653993613 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Say the probability of the sequence "Seismology is a ground breaking experience" is p , then what is $1000 * p$?

$$0.42 \times 0.43 \times 0.29 \times 0.48 \times 0.29 \times 0.23$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1.6

$$= 0.0016$$

$$= 1.6$$

Question Number : 287 Question Id : 640653993614 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the probability of the sequence "Seismology is earthquake study"?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-1

Question Number : 288 Question Id : 640653993615 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Suppose we use Top- k sampling,
with $k = 3$, what is the probability
of the predicted word equal to
“Seismology” given the “[Start]” token
as the initial input

$P(y_1 = \text{Seismology} | y_0 = [\text{Start}])?$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-1

Sub-Section Number :

Sub-Section Id :

Question Shuffling Allowed :

5

640653146275

No

Question Id : 640653993607 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (289 to 291)

Question Label : Comprehension

$$\begin{array}{r} 0.41 \\ 0.09 \\ 0.14 \\ \hline 0.64 \end{array}$$

$$\begin{array}{r} 0.2624 \\ \hline 0.41 \\ 0.64 \end{array}$$

Rajesh has created a GPT-like transformer model. However he doesn't have access to large compute infrastructure, so he has configured the original architecture in the following manner:

1. He has taken a vocabulary of size 1000 words/tokens.
2. Sequence length is 64.
3. Embedding dimension and d_{model} is 128.
4. There are only 4 transformer blocks (layers).
5. Each transformer block (layer) has only 4 attention heads.
6. FFN hidden layer size is 256.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 289 Question Id : 640653993608 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What will be the shape of positional embedding?

Options :

6406533355917. ✓ 64×128

6406533355918. ✗ 64×4

6406533355919. ✗ 12×128

6406533355920. ✗ 512×768

6406533355921. ✗ None of these.

$$d_g = \frac{d_{model}}{4} = \frac{128}{4} = 32$$
$$Tx \quad d$$
$$64 \times 128 =$$

Question Number : 290 Question Id : 640653993609 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What will be the shape of W_Q^i ?

Options :

6406533355922. ✗ 32×32

6406533355923. ✗ 64×32

6406533355924. ✗ 128×16

6406533355925. ✓ 128×32

6406533355926. ✗ 8×32

6406533355927. ✗ None of these.

$$d_{model} \times d_g$$
$$128 \times 32$$

Question Number : 291 Question Id : 640653993610 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What will be shape of the mask M ?

Options :

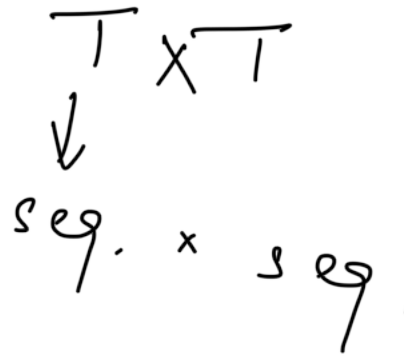
6406533355928. ✖ 16×16

6406533355929. ✖ 32×32

6406533355930. ✔ 64×64

6406533355931. ✖ 128×128

6406533355932. ✖ None of these.



Sub-Section Number :

6

Sub-Section Id :

640653146276

Question Shuffling Allowed :

Yes

Question Number : 292 Question Id : 640653993605 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

The input embeddings for the words “learning”, “brings” and “joy” are $h_1 = [0.5, 0.25, 1]$, $h_2 = [0.1, 0.25, 0]$, and $h_3 = [0.1, 0.1, 0.9]$, respectively. Note that the embeddings are row vectors. The projection matrices are as follows

$$W_Q = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} \quad W_K = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad W_V = \begin{bmatrix} 0 & 0 \\ -1 & -1 \\ 1 & 1 \end{bmatrix}$$

The following quantities are computed as

$$Q = HW_Q \quad K = HW_K \quad V = HW_V$$

Let e_j denote the unnormalized attention score, a_j denote the normalized attention score (ignore the scaling by $\sqrt{d_k}$) and z_j denote the linear combination of the value vectors for the j -th word.

Suppose the gradient vector $\frac{\partial L}{\partial z_3} = [1, 2]$, then what is the gradient vector $\frac{\partial L}{\partial e_3}$? Enter the sum of gradients.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal



Text Areas : PlainText

Possible Answers :

0

Sub-Section Number :

7

Sub-Section Id :

640653146277

Question Shuffling Allowed :

Yes

Question Number : 293 Question Id : 640653993606 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider the embedding vector for a word, $x = [0.1, 0.2, -0.3, 0.4]$. Suppose the word is at position 2 in the given sentence. Add the corresponding position embedding p to the word embedding to get h , i.e. the sum of the elements in $h = x + p$. Use the fixed-sinusoidal position embedding vector calculated using the formula given below

$$i = 0, 1$$

$$pos = 2$$
$$d_{model} = 128$$

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$i = 0 \rightarrow 2i \Rightarrow 0$$
$$\rightarrow 2i+1 \Rightarrow 1$$

$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$\sin\left(\frac{2}{10000^{2/128}}\right)$$

What is $h[0] + h[1]$ i.e. sum of first two elements of h ?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.75 to 0.85

$$= 0.9$$

$$0.1 + 0.9 = 1$$

$$\frac{2}{10000^{1/128}}$$

$$= \frac{2}{10000^{1/128}} = \cos(2)$$
$$= -0.4161$$

DLP

$$= -0.2$$

Section Id :

64065369324

Section Number :

15

Section type :

Online

Mandatory or Optional :

Mandatory

Number of Questions :

14

Number of Questions to be attempted :

14

Section Marks :

50

Display Number Panel :

Yes

Section Negative Marks :

0

$$1 - 0.2 = 0.8$$